

THE OHIO Naturalist

PUBLISHED BY
THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY

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Volume 1.

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No. 8, June 1901**

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A STUDY IN VARIATION ON THE WING OF THE HONEY BEE.

F. L. LANDACRE.

While working on the Honey Bee in the laboratory at the University it was observed that the number of hooks connecting the posterior wing with the anterior was not constant.

The query at once arose as to the amount of variation there might be between different bees in the same hive and also between different hives. Out of this grew a somewhat practical problem as to whether the increase in number of hooks was associated with a decrease in the size of the wing, or whether the increase in number of hooks also implied an increase in size of wing.

The wings of the bee are undoubtedly more efficient for being closely attached to each other. The life of the workers is so short, being only about three weeks, and their activity so great that any increase in efficiency, especially in the organs of flight, must have a very direct influence on the welfare of the whole swarm. So far as the well being of the swarm depends upon nutritive processes the efficiency of the hive is equal to the average efficiency of the workers. Now, if the increase in number of hooks and the consequent, firmer attachment of the wings is compensated for by a smaller wing, there is much less opportunity for the operation of natural selection on the individual bees than if the greater number of hooks is always associated with a broader or longer wing.

This selective process might occur either in the hive or between hives. If it occurs in the hive it would increase the efficiency of the hive somewhat; but if it occurs between hives it finally means the elimination of the weaker hive and the consequent increased efficiency of the species.

In order to find out the real conditions, one of the students, Mr. J. N. Frank, took twenty-five workers from each of four hives and counted the number of hooks on each wing, right and left, and also measured the width of the anterior and posterior wings on each side, The width only was taken

on account of the difficulty in finding a good point at the base of the wing from which to measure the length. The results are so uniform that the width probably gives sufficient data from which to draw conclusions.

Of the four hives studied, numbers one and two were very weak. Number three was a strong hive which made forty (40) pounds of extra honey in the summer of 1900. Number four was weaker than number three and made only ten (10) pounds of extra honey.

The complete measurements are too long to give in detail, and the averages only will be offered here.

AVERAGES OF TWENTY-FIVE MEASUREMENTS FOR EACH
HIVE TAKEN WITH AN EYE-PIECE MICROMETER, EXPRESSED IN
MM.

HIVE NUMBER ONE.

Average Number of Hooks.		Average Width of Wing.			
Right wing.	Left wing.	Right wing.		Left wing.	
		Ant.	Pos.	Ant.	Pos.
21.3	20.9	4.21	3.61	4.28	3.55

HIVE NUMBER TWO.

Average Number of Hooks.		Average Width of Wing.			
Right wing.	Left wing.	Right wing.		Left wing.	
		Ant.	Pos.	Ant.	Pos.
19.2	18.8	4.14	3.48	4.16	3.48

HIVE NUMBER THREE.

Average Number of Hooks.		Average Width of Wing.			
Right wing.	Left wing.	Right wing.		Left wing.	
		Ant.	Pos.	Ant.	Pos.
21.0	21.0	4.06	3.48	4.07	3.50

HIVE NUMBER FOUR.

Average Number of Hooks.		Average Width of Wing.			
--------------------------	--	------------------------	--	--	--

Right wing.	Left wing.	Right wing.		Left wing.	
		Ant.	Pos.	Ant.	Pos.
19.6	19.6	4.09	3.47	4.03	3.41

As to the first query concerning the individual variations in a single hive the complete table shows that No. 1 varies from 18-21 hooks, No. 2 from 17-21, No. 3 from 18-23 and No. 4 from 17-21. The right wing is taken as the standard, and the most active hive, No. 3. shows the greatest individual variation. One bee in this hive had only sixteen hooks, the remaining three being straight spines, showing how the hooks have been modified from ordinary hairs. This reversion occurred on three separate wings, in No. 2 one hook on each wing being straight.

As to the relation between the number of hooks and the width of the wing the averages are very definite. Taking hives number one and two from the same apiary, it will be seen that the increase in number of hooks goes with the increase in width of wing. The same relation is shown by hives numbers three and four from another apiary in the case of the posterior portion of the right wing and in both anterior and posterior portions of left wing.

The results are not conclusive as to the relative efficiency of different hives because there are so many conditions entering into the production of large quantities of honey. The number of bees, the care during the winter, the age of the queen, the number of swarms produced, and several other factors would have to be taken into consideration.

The differences in the right and left wings in the bees of the same hive is marked. The right wing has the larger number of hooks, but the left wing is the broader. In hive number one the average number of hooks in the right wing is 21.3, left 20.9; but the anterior wing on the right side is 4.21 mm., while the left anterior wing is 4.28; that is, there is a compensation for the reduced number of hooks in the increased width of the wing. This is true of the first three hives. In the fourth hive there is a slight advantage in favor of the right wing.

The following general conclusions may be drawn from these measurements:

(a) There is a variation in the number of hooks in a given hive ranging between 17 and 23.

(*b*) The difference in the number of hooks in the right and left wing is compensated for in a given hive by the increased size of the wing. The right and left wings are in physiological equilibrium.

(*c*) In different hives the increase in the number of hooks is accompanied by an increase in width of wing; that is, the variation is emphasized so that selection would work much more effectively; while in the individual, where—if selection operated on account of this variation—it would have to be between different wings of the same bee, the variation is eliminated.

TWELVE PLANTS ADDITIONAL TO THE OHIO LIST.

W. A. KELLERMAN.

The species named below have not heretofore been recorded as a part of the Ohio flora. The first collector and locality are given for each of the listed species. The serial number prefixed to each name indicates where in the Fourth State Catalogue the species should be inserted.

123a *Sorghum vulgare* Pers. Occasionally escaped.

270b *Secale cereale* L. Rye. Occasionally escaped.

781b *Dianthus barbatus* L. Sweet William. Escaped. Painesville. Otto Hacker.

(1042a *Crataegus polybracteata* Ashe. Reported previously, but without locality. Franklin, Hocking, and Summit Counties; W. A. Kellerman.)

1042b *Crataegus pruinosa* Wendl. Logan County; W. A. Kellerman.

1042c *Crataegus succulenta*. Franklin, Fairfield, Knox, Belmont, Summit, Ottawa, Union, Ross, Carroll, Shelby and Lucas Counties; W. A. Kellerman.

1526b *Teucrium occidentale* Gr. Hairy Germander. "Ohio", Riddell, 1834, (Bull. Torr. Club, 28:170); Reservoir Park, Perry Co., W. A. Kellerman.

1529a *Scutellaria serrata* Andr. Showy skullcap. Rio Grande. Gallia County; Ruth E. Brockett.

1605a *Solanum tuberosum* L. Potato. Escaped.

1609c *Petunia violacea* Lindl. Occasionally escaped.

1709a *Viburnum molle* Mx. Soft-leaf Arrow-wood. Scioto County; W. A. Kellerman.

1714a *Linnaea borealis* L. Twin-flower. Canton, Stark County; Mrs. Theano W. Case.

1986a *Chrysanthemum indicum* Hortorum. Escaped. Adams County; W. A. Kellerman.

THE FOOD HABITS OF SOME APHROPHORA LARVAE.

E. D. BALL.

The larvae of all the American species of the Family Cercopidae as far as known envelope themselves in a frothy mass. Contrary to popular opinion and to most of the published accounts this froth does not issue as bubbles from the body of the insect, but is made by pushing the tip of the abdomen up out of the froth and grasping, with the anal appendages, a bubble of air and bringing it down and releasing it within a liquid film. This liquid film is simply the excretion from the alimentary canal of the sap which is imbibed by these insects in large quantities. This copious liquid excretion is a common occurrence in other families of the Homoptera. In the Plant Lice (Aphidae) it gathers in drops and is called "Honey Dew." The Leaf Hoppers and Tree Hoppers expel a clear liquid with some force. In some species this is in sufficient amount so that when the insects are numerous the foliage may drip, producing the "Weeping trees" of the Southern States.

This process of froth making in the Cercopidae was discovered and first correctly described by Professor E. S. Morse, of Salem, Mass., and published many years ago in his Elementary Zoology.^[1] His observations were probably made on the larvae of *A. spumarius* which belongs to the genus *Philaenus* as now recognized.

In the genus *Aphrophora* as now limited little is known of the food habits of the larvae. One species (*A. 4-notata*) has been found on various plants and shrubs. The remaining three eastern species, which belong to a different group and are of some shade of brownish testaceous, have been given as feeding on pines in the adult state by various authors. Dr. Fitch has described the larvae of one of these (*A. parallela*, Fig. 4, Plate 10) as forming frothy masses on the tips of pine twigs, and in the Nat'l Museum Coll. are some *Aphrophora* larvae labeled "Pa. On Pine, July 7," that undoubtedly belong to this species leaving little room to doubt the correctness of Fitch's determination.

There are two species belonging to the *parallela* group occurring in the Rocky Mountain region both found in the adult stage on pines. Of one of these (*A. permutata*, Fig. 1, 2 and 3, Plate 10) larvae were found in abundance on two different plants *Chrysopsis villosa* and *Lupinus sp.* Both of these plants grow in clumps and it was always down in the bases of these clumps, some of them often down below the surface of the ground among the roots, that the larvae were found. Often ten or fifteen would be found in a single clump their united froth masses, held up by the coarse stems, reaching a diameter of two inches or more.

The larvae were found in these clumps from late in May until the first week in July in the foot hills, and higher up in the mountains they were just beginning to emerge July 20th. When ready to emerge they climb up a stem during the night far enough to free themselves from the froth and as soon as the sun strikes then in the morning they burst their pupal skins and an hour later they are ready to fly up to the pine trees where their color admirably protects them.

Although both these plants grow very commonly over a wide extent of territory the *Aphrophora* larvae have never been found on them except where they were within a short distance of a pine tree. At first sight it would seem probable that the eggs were deposited in the twigs of the pines, and that the young larvae dropped to the ground, and from there sought out a food plant, as is the case in some Cicadas. But as numerous larvae were found in positions practically inaccessible to any such means of distribution—such as on the opposite side of a sharp ledge of rocks, across a bramble thicket, or even on plants growing in the crevices of bare rocks at a considerable distance above any pines—it seemed nearly certain that the adults must fly back to the plants to deposit their eggs.

It will be interesting to discover whether a similar food habit occurs in any of the Eastern members of the genus or whether this is peculiar to the western species. It seems possible that original pine-inhabiting species finding themselves unable to maintain their froth masses in their exposed positions on pine branches in such a dry atmosphere were compelled to seek moister conditions such as are afforded by the shade and contact with the earth under these bushy plants.

EXPLANATION OF PLATE.—Fig. 1. *Aphrophora permutata* Uhl. Adult. X About 7. Common from the Rocky Mountains to the Pacific. 1a—Side view of head, showing profile of face.

Fig. 2. Pupa of above. 2a—Side view of head of pupa.

Fig. 3. Semi-diagrammatic cross-section of a clump of *Chrysopsis villosa* to show relative location of larvae with relation to the froth mass and the surface of the earth.

Fig. 4. *Aphrophora parallela* Say Adult. X About 7. Eastern U. S. to Ohio and Mich. 4a—profile of head of same.

FOOTNOTES:

[\[1\]](#) For a detailed account of this process see Prof. Morse's article "A Bubble-blowing Insect." Pop. Sc. Monthly, May, 1900.

Plate 10.

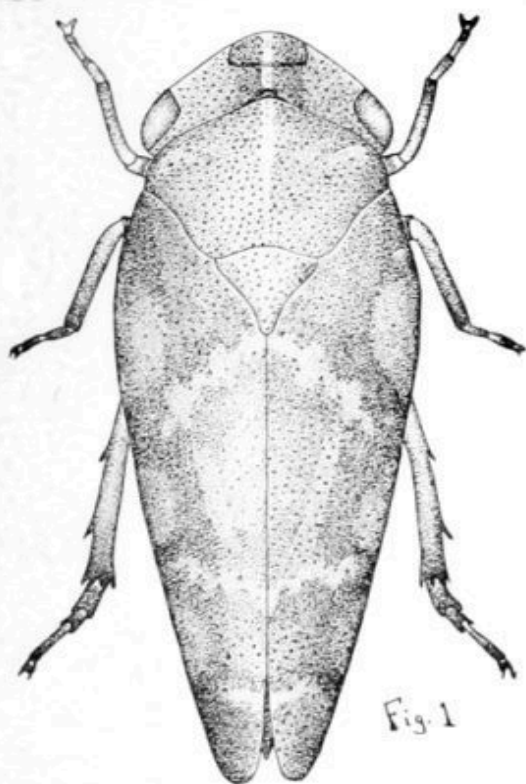


Fig. 1

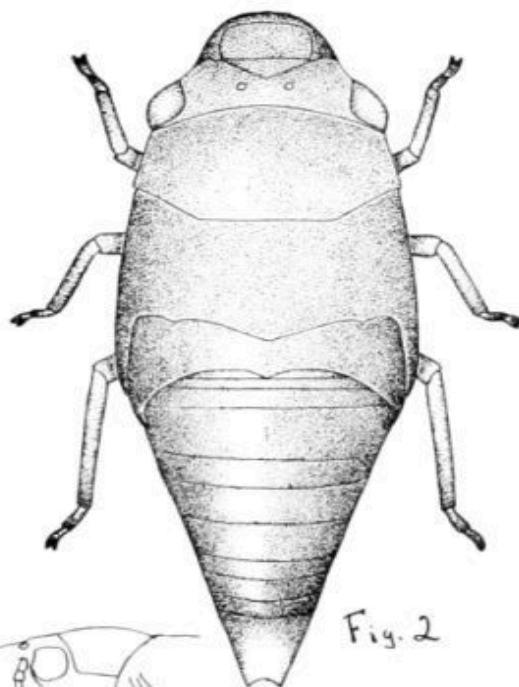


Fig. 2

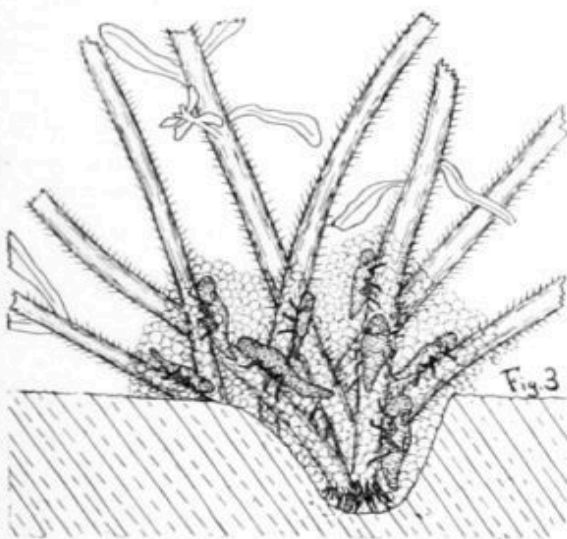
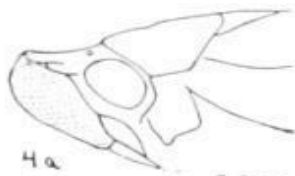
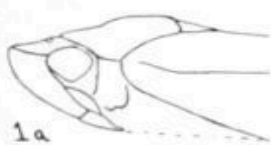


Fig. 3

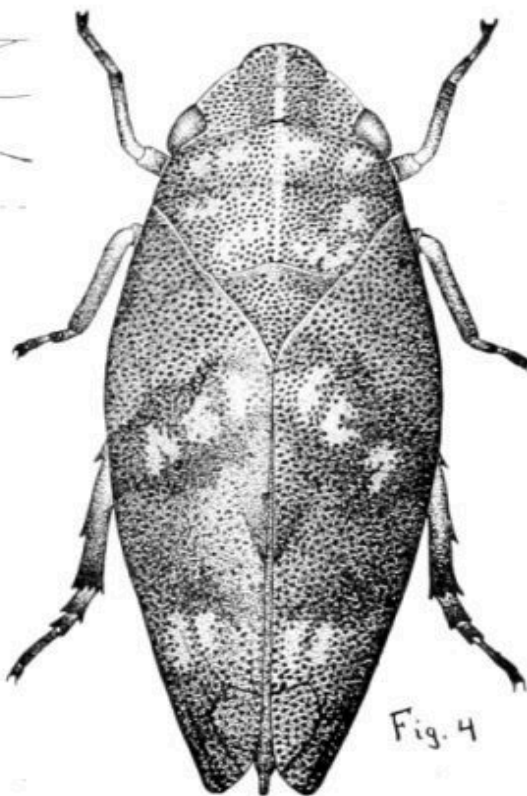


Fig. 4

E. D. M. N. Ball del.

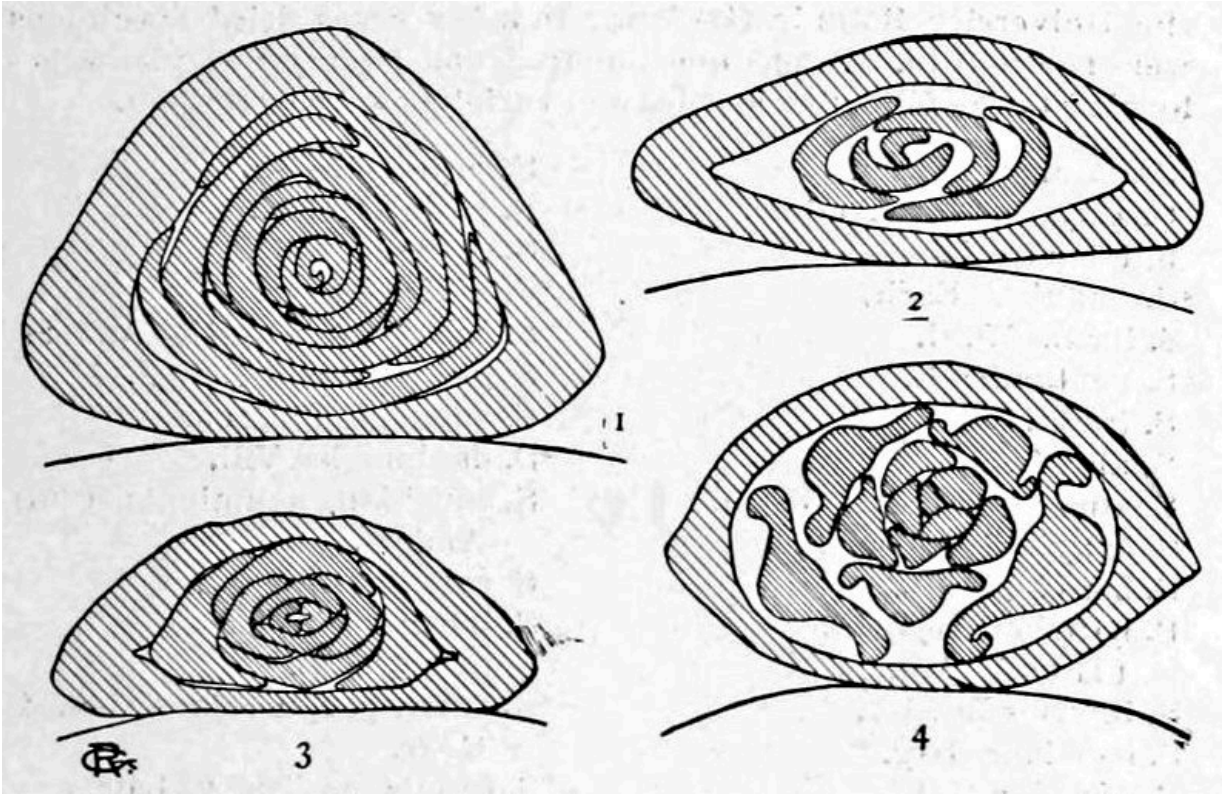
BALL ON APHROPHORA LARVAE.

THE VERNATION OF SALIX.

ROBERT F. GRIGGS.

Most of the manuals are entirely silent regarding the vernation of the Willows. Sargent^[2] describes their leaves as “variously folded in the bud” and under different species gives them as: involute, revolute, convolute, and even conduplicate in the bud. The fact that he gives two species, closely related and difficult to distinguish, at the time the buds open (*Salix nigra* and *S. amygdaloides*), as having involute and revolute vernations, led me to take up the matter to see if a key for their identification from bud characters, could be constructed.

Not only did I find that they were not involute and revolute respectively; but that they were neither involute nor revolute, but both imbricate. On examining other species the same thing was found. The only exceptions to the true imbricate vernation found are represented in figures 3 and 4. The section *Purpureae*, on account of its tendency to have opposite rather than alternate leaves, often forms such decussate buds as are shown in figure 3. In *Salix incana* Schrenk, a species whose leaves at maturity have revolute margins, the leaves have a greater or less tendency to roll backwards in the bud. The most extreme case found is shown in figure 4. Others from the same twig could be shown where the leaves show only the slightest tendency to be revolute. If we may consider that this backward turning is merely a character of the mature leaf manifesting itself in the bud it is evident that there is here no revolute vernation but that it is really imbricate.



Salix buds

The other buds examined vary from the form represented in figure 1 where the whole interior of the bud is taken up with the closely packed leaves, to that shown in figure 2 where there are a few leaves with a great deal of wool.

Species like *Salix fragilis* L. whose leaves are glabrous when they unfold have buds like the former while species like *Salix discolor* Muhl., with leaves excessively wooly when they unfold, are like the latter. As there are all intergradations between these two kinds of leaves, there is naturally a series of buds between these two as extremes. While further investigation is necessary before we would be warranted in declaring that the vernation of the whole genus is imbricate; yet the fact that specimens of thirty-four species and varieties, taken from thirteen of the nineteen sections given by Andersson in DeCandolle's prodromus, have their leaves imbricated in the buds would seem to establish a presumption in favor of such a view.

The buds examined were soaked in 70% alcohol and free-hand sections cut and mounted in balsam. On account of scarcity of material, the buds of several species were not sectioned but dissected on the growing plant. Such

are marked with an asterisk (*). As far as possible living material was taken, mostly from native plants. Those species not native were studied from specimens growing in the University Botanic Garden. In a few cases dried specimens were resorted to. About one hundred and twenty-five plants belonging to the following species and varieties were examined.

Salix nigra Marsh.	S. sericea Marsh.
S. amygdaloides Anders.	S. petiolaris Sm.
S. triandra L.	S. cordata Muhl.
*S. undulata Ehrh.	*S. cordata x sericea.
S. lucida Muhl.	S. cordata var. vestita Anders.
S. pentandra L.	S. glaucophylla Bebb.
S. fragilis L.	S. adenophylla Hook.
S. alba L.	*S. daphnoides Vill.
S. alba vitellina (L.) Koch.	S. smithiana acuminata (Sm) Anders.
S. babylonica x fragilis.	S. candida Fluegge.
S. babylonica L.	S. incana Schrenk.
S. babylonica japonica (Thumb) Anders.	S. purpurea L.
S. interior Rowlee.	*S. rubra purpureoides Gen. & Godr.
S. bebbiana Sarg.	*S. candicans Gen. & Godr.
S. discolor Muhl.	*S. laurifolia Gen. & Godr.
S. myrtilloides L.	*S. sieboldii Gen & Godr.
S. humilis Marsh.	
S. tristis Ait.	

EXPLANATION OF THE FIGURES.—Fig. 1. *Salix fragilis* L. Bausch and Lomb obj. $\frac{2}{3}$ oc. 2.

Fig. 2. *S. discolor* Muhl. B. & L. obj. $\frac{2}{3}$ oc. 2.

Fig. 3. *S. purpurea* L. B. & L. obj. $\frac{2}{3}$ oc. 2.

Fig. 4. *S. incana* Schrenk. B. & L. obj. $\frac{2}{3}$ oc. 1.

The figures were drawn with an abbe camera lucida and reduced to $\frac{2}{5}$ of their original size.

FOOTNOTES:

[\[2\]](#) Sargent: Silva of North America 9: 95.

OHIO REPTILES IN THE OHIO STATE UNIVERSITY ZOOLOGICAL MUSEUM.

MAX MORSE.

Fam. IGUANIDÆ.

Sceloporous undulatus undulatus (Latr.). Sugar Grove.

Fam. ANGUIDÆ.

Ophisaurus ventralis (Linn.). No. 57 has the following note in the accession catalogue:—"Donated by Dr. N. S. Townshend. Said by him to have been taken on the University farm."

Fam. SCINCIDÆ.

Eumeces quinquelineatus (Linn.). Columbus.

Fam. COLUBRIDÆ.

Carphophiops amoenus (Say.). This specimen is marked *C. helenae* Kenn. The internasal scutae are wanting. The specimen is from Meigs Co. Another specimen is in the collection but without label.

Diadophis punctatus (L.) Sugar Grove.

Heterodon platyrhinus Lat. Cedar Point. Both the spotted and the black forms occur on Cedar Point.

Liopeltis vernalis (DeKay.). Sandusky and Columbus.

Zamenis constrictor (Linn.). Hocking County.

Coluber vulpinus (B. and G.). The range of the fox snake, as given by Cope (Rept. U. S. Nat. Mus., '98, p. 832) is "over the northwest of the Eastern district, not being known from east of Illinois***." Specimens are taken from Cedar Point and vicinity nearly every summer. Those in the collection are from Castalia and Cedar Point.

Coluber obsoletus obsoletus Say. Columbus.

Osceola doliata triangula (Boie.). Columbus and London.

Natrix fasciata fasciata (Linn.). Warren County.

Natrix fasciata sipedon (Linn.). This is the common “water snake” of central Ohio. Specimens are from Sandusky and Columbus.

Natrix fasciata erythrogaster (Shaw.) Put-in-Bay. Among the islands of Lake Erie this seems to be the prevailing form. One young *Natrix* was taken during the summer of 1900 which resembled *N. f. sipedon* L., but aside from this all other forms were *erythrogaster*. It may be possible that the young of these two sub-species are not distinguishable—the differences arising later.

Natrix leberis (Linn.). Columbus and Sandusky.

Natrix kirtlandii (Kenn.). New London and Sugar Grove.

Storeria dekayi (Holb.). Columbus.

Storeria occipitomaculata (Storer). Sugar Grove and Kent.

Eutaenia sirtalis sirtalis (Linn.). Columbus.

Eutaenia sirtalis ordinata (Linn.). Columbus.

Eutaenia sirtalis obscura Cope. Cedar Point and Columbus.

Fam. CROTALIDÆ.

Ancistrodon contortrix (Linn.). Sugar Grove and Knox County.

Sistrurus catenatus catenatus (Raf.). Urbana.

Fam. TRIONYCHIDÆ.

Aspidonectes spinifer (LeSueur.). Columbus.

Fam. KINOSTERNIDÆ.

Aromochelys odoratus (Lat.). Columbus and Cedar Point. In May and June, numbers of this turtle have been taken in the sand on Cedar Point, while depositing their eggs.

Fam. EMYDIDÆ.

Graptemys geographicus (LeSueur.). Columbus, also taken at Sandusky.

Chrysemys marginata (Agassiz). Columbus and Cedar Point. The species *picta* does not occur in central Ohio—at least west of Licking Reservoir. One specimen has the normal three dorsal plates broken up into six which alternate with each other, three being on either side of the median line.

Clemmys guttatus (Sch.). Columbus and Licking Reservoir.

Emydoidea blandingi Holb. Columbus and Sandusky.

Terrapene carolina (Linn.). Very numerous at Sugar Grove. Found in sand on Cedar Point.

SUMMARY FOR REPTILES:—Families 8; genera 22; species 30.

A PRESERVING BOX FOR PLANTS.

EDO CLAASSEN.

As the time for botanists has arrived when they will depart for some time from their work at home and walk over fields and into the forests to collect plants and flowers new to them, I have thought it would be interesting and useful to describe a box in which they may preserve for several days, the collected plants and keep them from shriveling, particularly if the same are quite large, and exceed in size the usual small collecting box. As I had one made to order and know by experience the valuable service it did me, I do not hesitate to recommend it highly. It is well known that many druggists buy their glycerine and castor oil in five gallon cans, for which, when empty, they have no further use. The botanist, therefore, may go to such a druggist, procure two of the above cans, if possible of heavy tin and with flat sides, have the tinsmith take off their upper parts and solder the cans together, after having cut out of each of them a rectangular piece as long and wide as necessary to give room for a door and after having trimmed any inside edges. The door is then made from the two pieces cut out, (or from a new piece) with the addition of several strips of tin, so that it may overlap and close tightly, and of the necessary hinges and hasp to open and fasten the door. One of the original wire handles of the cans is fastened in a similar manner as before on the top of the box and the preserving box is ready for use, as soon as it had received two coatings of asphaltum varnish inside and two of paint outside. Any vessel of suitable size and containing water should then be put into the box, which will furnish the moisture for the roots or the lower ends of the plants and at the same time for the air surrounding these. The dimensions of the box in question can easily be determined by the botanist himself, but for those not wishing to do so, I may be allowed to add, that the length of the box should be about twenty-five inches, the original width of the cans remaining unchanged. The door should commence at about three inches from the bottom, reach up to two or two and one-half inches from the top and have a width of six or six and one-half inches.

Cleveland, Ohio.

OHIO TUMBLEWEEDS.

JOHN H. SCHAFFNER.

Tumbleweeds are characteristic of wind-swept plains and dry prairies. As the forests are rapidly disappearing, the conditions in Ohio are becoming very favorable for the introduction and development of such forms of vegetation. A few species are already abundant and some like *Amaranthus graecizans* appear to flourish better than on the prairies of the interior. The past summer a number of cornfields about Columbus were covered with very large tumbleweeds and during the winter a number of hedgerows were filled with them, presenting an appearance quite as striking as anything the writer has seen along this line.

The following is a list of the Ohio plants which may develop as tumbleweeds. Those with a question mark have not been seen by the writer to act as tumbleweeds and a few are given on the authority of Dr. W. J. Beal.

ANNUAL TUMBLEWEEDS.

1. *Cycloloma atriplicifolium* (Spreng.) Coult.
2. *Salsola tragus* L.
3. *Amaranthus graecizans* L.
4. *Lepidium apetalum* Willd. Beal.
5. *Trifolium procumbens* L. Beal.
6. *Onagra biennis* (L.) Scop. Beal.

TUMBLE-GRASSES.

7. *Panicum capillare* L.
8. *Panicum flexile* (Gattg.) Scrib.
9. *Eragrostis pectinacea* (Mx.) Steudl.
10. *Eragrostis trichodes* (Nutt.) Nash.
11. *Eragrostis capillaris* (L.) Nees. ?
12. *Eragrostis frankii* Steud. ?
13. *Eragrostis purshii* Schrad. ?
14. *Agrostis hyemalis* (Walt.) B. S. P.

PERENNIAL TUMBLEWEEDS.

15. *Baptisia tinctoria* (L.) R. Br.

MEETINGS OF THE BIOLOGICAL CLUB.

MAY MEETING.

The Biological Club met in Zoological Lecture Room May 6, 1901.

Professor Schaffner reported that the committee appointed to consider the disposition to make of exchanges, had had a meeting and appointed Professor Osborn to consider the matter further.

Professor Landacre gave a paper entitled A Study of *Passalus Cornutus*. He gave the more important conclusions he had arrived at, after an extended study of the muscular and skeletal systems of that beetle.

Mr. Griggs read a paper on Vernalion in the Willows.

Moulds and other Fungi Injurious to Foods was the title of a paper given by Miss Mary Dresbach. She gave a list of fungi found on food products.

In discussing this paper Professor Schaffner said the moulds are of public interest and many important results may be expected from an extended study of them. Professor Kellerman said that moulds are an important factor to guard against in canning fruit. It would be a great step in advance if fruit could be canned and kept without its being cooked beforehand.

Miss Elma Perry gave a list of the edible fungi of Ohio. So far 200 species have been recorded and there are no doubt many yet to add.

Under the head of personal observations Prof. Kellerman showed some interesting variations in our common anemone. A paper on this subject appeared in the May number of *THE OHIO NATURALIST*.

JUNE MEETING.

The June meeting of the club was held in Zoological Lecture Room on the evening of the third.

The editor-in-chief of *THE OHIO NATURALIST*, Professor Schaffner, gave a financial statement for the year.

The Secretary read a communication from Professor Kellerman. This communication was concerned with the past and future of THE OHIO NATURALIST and an adjourned meeting was voted for its consideration. The following invitation was received from Professor W. D. Gibbs, Secretary of the Omega Chapter of the Society of Sigma Xi:

The Omega Chapter of the Society of Sigma Xi cordially invites the members of the Biological Club to be present at the final meeting of the Chapter, in the Physical Lecture Room, Saturday June 8th, at 11 o'clock a. m. to hear a lecture by Professor Charles F. Mabery on the subject: "The Petroleum Industry: Its Rapid Expansion and Future Promise."

Dr. Bownocker delivered an instructive paper on "Oil and Gas in Southeastern Ohio." The first oil well in that region was drilled about 1860 at Maxburg in Washington County. The wells in this region when first put down oftentimes yield 500 barrels or more daily but they soon decrease until the product of the same well may be only a few barrels each day. They are long-lived however and wells drilled in the Sixties are still producing. The oil in Southeastern Ohio has its origin in the various sands and therefore differs from that of Northwestern Ohio where all the oil is of limestone origin.

The speaker explained the nature of the country, especially in reference to the arches where practically all the oil is located.

H. S. Houghton not being present, Professor Landacre gave a short outline of the subject "A Study of the Muscular and Skeletal Structures in the Head of a 14 mm. Salamander."

A. F. Conradi read a short paper on the subject "A Study of the Cecidomyidae and their effects upon Vegetation."

The motion was made and carried that when we adjourn we adjourn to meet in the Zoological Lecture Room Friday evening, June 7th, at 4 o'clock, for the purpose of considering matters connected with the THE OHIO NATURALIST.

JAMES S. HINE, Secretary.

With this issue THE OHIO NATURALIST completes its first year. The Editors to be chosen for 1901-2 will continue the Journal along the same lines and we trust that those interested in the natural history of Ohio, as well as others, will continue to give their encouragement and financial support.

A table of contents and a title-page of Vol. 1 will be sent out with the first number of Vol. 2.

JOHN H. SCHAFFNER.

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Transcriber's Notes

pg 124 Changed Semi-diagramatic to: Semi-diagrammatic
pg 126 Assumed reduced to 2-5 to mean: $\frac{2}{5}$
pg 128 Removed duplicate word of: vessel of of suitable
pg 130 Changed the Ohio Naturalist. to: The Ohio Naturalist
pg 131 Added comma after: Engineering, Pharmacy
pg 131 Added period after: Keys simple.
pg 132 Added period after: David S. Jordan M. D
Table of Contents capitalization and punctuation left as written-
does not match chapter headings

*** END OF THE PROJECT GUTENBERG EBOOK THE OHIO
NATURALIST, VOL. 1, NO. 8, JUNE 1901 ***

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